

```

#include <stdio.h>
#include <stdlib.h>
void Allpairs(int **graph, int n)
{
    int i, j, k;

    for (k = 0; k < n; k++)
    {
        for (i = 0; i < n; i++)
        {
            for (j = 0; j < n; j++)
            {
                if (graph[i][j] > graph[i][k] + graph[k][j])
                    graph[i][j] = graph[i][k] + graph[k][j];
            }
        }
    }
}

int main(void)
{
    int n, i, j;

    printf("Enter the number of vertices: ");

    scanf("%d", &n);

    int **graph = (int **)malloc((long unsigned) n * sizeof(int *));

    for (i = 0; i < n; i++)
    {
        graph[i] = (int *)malloc((long unsigned) n * sizeof(int));
    }
}

```

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for (i = 0; i < n; i++)
{
    for (j = 0; j < n; j++)
    {
        if (i == j)
            graph[i][j] = 0;
        else
            graph[i][j] = 100;
    }
}

printf("Enter the edges: \n");

for (i = 0; i < n; i++)
{
    for (j = 0; j < n; j++)
    {
        printf("[%d][%d]: ", i, j);
        scanf("%d", &graph[i][j]);
    }
}

printf("The original graph is:\n");

for (i = 0; i < n; i++)
{
    for (j = 0; j < n; j++)
    {
        printf("%d ", graph[i][j]);
    }

    printf("\n");
}

Allpairs(graph, n);

```

```
printf("The shortest path matrix is:\n");  
  
for (i = 0; i < n; i++)  
{  
    for (j = 0; j < n; j++)  
    {  
        printf("%d ", graph[i][j]);  
    }  
    printf("\n");  
}  
  
return 0;  
}
```